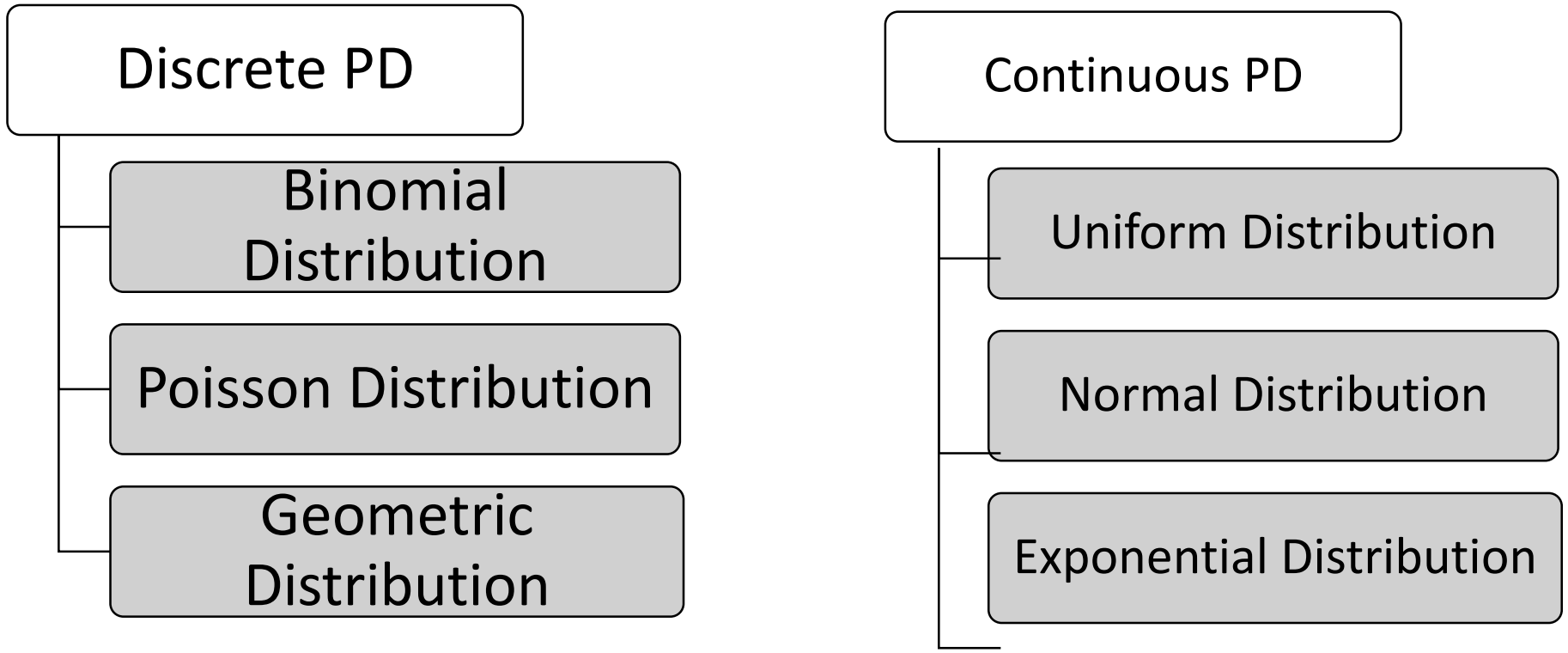
Probability Distribution (2)

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Probability Distribution



Binomial Distribution

- First we need to learn about Bernoulli trial.

- Suppose an experiment has only two outcomes
  a. Success
  b. Failure

Bernoulli variable

$$X = \begin{cases} 1; if\ succsess\ occurs \\ 0; if\ failure\ occurs \end{cases}$$

Binomial Distribution

- "**n**" independent Bernoulli trials are performed
- Each trial has two outcomes: Success and Failure
- Probability of success p and probability of failure $(1 - p)$

Binomial Distribution

- Probability mass function of binomial distribution,

$$p(x) = {}^{n}C_{x}\, p^{x}\, (1-p)^{n-x}; x = 0,1,2,3, \dots, n$$

$$X \sim binomial\ (n, p)$$

"n" and "p" are Parameter

Binomial Distribution

- Mean of the binomial distribution

$$\mu = E(X) = \sum x\, p(x) = np$$

- Variance of the binomial distribution

$$\sigma^2 = E(X^2) - \big(E(X)\big)^2 = np(1-p)$$

Binomial Distribution

A fair coin is tossed 4 times. Find the probability of (a) exactly two heads,(b) at least 4 heads,(c) at most 2 head,(d) no heads, e) Find the mean and variance of that distribution.

Solution: Let the number of heads be random variate X which can take values 0, 1,2,3, and 4. X is a binomial variate with probability $\frac{1}{2}$ and $n = 4$.

Binomial Distribution

(a) exactly two heads

$$p(x) = {}^{n}C_x \, p^x \, (1-p)^{n-x}$$

$$p(X=2) = {}^{4}C_2 \left(\frac{1}{2}\right)^2 \left(1-\frac{1}{2}\right)^{4-2}$$

$$\therefore p(X=2) =$$

(b) at least 3 heads

$$p(x) = {}^{n}C_x \, p^x \, (1-p)^{n-x}$$

$$p(X \geq 3) = P(X=3) + P(X=4)$$

$$\therefore p(X=3) = {}^{4}C_3 \left(\frac{1}{2}\right)^3 \left(1-\frac{1}{2}\right)^{4-3}$$

$$\therefore p(X=4) = {}^{4}C_4 \left(\frac{1}{2}\right)^4 \left(1-\frac{1}{2}\right)^{4-4}$$

$$\therefore P(X=3) + P(X=4) =$$

Binomial Distribution

(c) at most 2 head

$$p(x) = {}^{n}C_x \, p^x \, (1-p)^{n-x}$$

$$p(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$\therefore p(X \leq 2) =$$

(d) no heads

$$p(x) = {}^{n}C_x \, p^x \, (1-p)^{n-x}$$

$$p(X = 0) = {}^{4}C_0 \left(\frac{1}{2}\right)^0 \left(1 - \frac{1}{2}\right)^{4-0}$$

$$\therefore p(X = 0) =$$

Binomial Distribution

- $Mean = np = 4 \times \frac{1}{2} = 2$

$Mean = np$

- $Variance = np(1-p) = 4 \times \frac{1}{2} \times \frac{1}{2} = 1$

$Variance = np(1-p)$

Binomial Distribution

- In a class, the probability that a student will be pass $\frac{4}{5}$. Among the 4 students in that class, what is the probability that

  a. All the four students will pass (Ans: ?)
  b. No students will pass (Ans: ?)
  c. Exactly one student will pass. (Ans:?)

Binomial Distribution

- The probability that a blood cancer patient who are under treatment will survive is 0.2. What is the probability that out of 6 patients

  a. 4 or more will survive (Ans: 0.01696)
  b. exactly 2 workers will survive? (Ans:0.24576)

Binomial Distribution

- The mean and standard deviation of binomial variable are 20 and 2, respectively. Find the parameters and probability function of the distribution.

$$Ans: n = 25 \; and \; p = 0.8$$

$$p(x) = {}^{25}C_x \; 0.8^x \; (1 - 0.8)^{25-x}$$

Poisson Distribution

- When data represents the number of occurrence of a specific event in a fixed period of time.

Poisson Distribution

Binomial dist. describe the distribution of binary data from **finite** sample

- When data represents the number of occurrence of a specific event in a fixed period of time.

Poisson dist. describe the distribution of binary data from **infinite** sample ($n \geq 100$)

- For example,
- "The number of call arrive at a helpline center in a given day"
- "The number of traffic accidents during a given time day"

Poisson Distribution

- The probability mass function of Poisson distribution:

$\lambda = np$

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; x = 0,1,2, \ldots$$

$Mean, E(X) = \lambda = np$
$Variance, Var(X) = \lambda = np$

$X \sim Poisson\ (\lambda)$

Parameter of Poisson dist.

Poisson Distribution

- Suppose an electrical manufacturing company produces 18 defective items on average in a day. What is the probability that in a given day there will be

  a. 10 defective items.
  b. 8 defective items
  c. At least 2 defective items.
  d. More than 10 but less than 15 defective items.

Poisson Distribution

a) 10 defective items

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; \lambda = 18$$

$$P(X = 10) = \frac{e^{-18}18^{10}}{10!}$$

$$P(X = 10) = 0.01498$$

b) 8 defective items

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; \lambda = 18$$

$$P(X = 8) = \frac{e^{-18}18^{8}}{8!}$$

$$P(X = 8) = ???$$

Poisson Distribution

c) At least 2 defective items

$\lambda = 18$

$P(X \geq 2)$ → $P(X \geq 3) = 1 - P(X < 2)$ → $P(X \geq 2) = 1$

$P(X = 0) + P(X = 1)$

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

Poisson Distribution

c) > 10 but < 15 defective items

$\lambda = 20$

$P(10 < X < 15)$

$P(10 < X < 15) =?$

$P(X = 11) + P(X = 12) + P(X = 13) + P(X = 14)$

$$P(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

Poisson Distribution

- The average number of errors on a page of a certain magazine is 0.2. What is the probability that the next page (or a randomly selected page) you read contains

  a. 0 (zero) error? (Ans: 0.8187)
  b. 2 or more errors? (Ans: 0.01756)
  c. What is the average error per page? (Ans: 0.2)
  d. Also, find standard deviation of the number of errors. (Ans: 0.45)

Geometric Distribution

- When we want to know how many how many trials required before achieving the first success in repeated Bernoulli trials, we may used geometric probability distribution.

Geometric Distribution

- Toss a coin and the event is “Head”
- First trial = Tail
- Second trial = Tail
- Third trial = Tail
- Fourth trial = Head

Geometric Distribution

- The probability mass function for geometric distribution

$$p(x) = p(1-p)^{x-1} \; ; x = 1,2, \ldots$$

- $Mean = \frac{1}{p}$

- $Variance = \frac{1-p}{p^2}$

Geometric Distribution

- A person wants to get a CNG that charges bills on a meter. . If the probability of getting meter charge CNG is $\frac{1}{5}$.

a. What is the probability that the fourth CNG is a meter charged?

b. Find the mean number of CNG to get a meter charged.

Geometric Distribution

$P(X = 4)$

$$P(X = 4) = \frac{1}{5} \times \left(\frac{4}{5}\right)^{4-1}$$

$$P(X = 4) = \frac{1}{5} \times \left(\frac{4}{5}\right)^{3} = 0.1024$$

$Mean$

$$Mean = \frac{1}{p}$$

$$Mean = \frac{1}{\frac{1}{5}}$$

$$Mean = 5$$

Thank You